Triangles & Congruence Error Analysis

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Triangle angle sum

ORIGINAL PROBLEM

The angles of a triangle measure 2x°, 3x° and 4x°. Find x.

INCORRECT WORKED SOLUTION

2x + 3x + 4x = 360

9x = 360

x = 40

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

2. Exterior angle

ORIGINAL PROBLEM

An exterior angle of a triangle measures 120°. The two remote interior angles measure x° and 2x°. Find x.

INCORRECT WORKED SOLUTION

x + 2x + 120 = 180

3x = 60

x = 20

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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3. Isosceles triangle

ORIGINAL PROBLEM

An isosceles triangle has a vertex angle of 40°. Find each base angle.

INCORRECT WORKED SOLUTION

180° − 40° = 140°

Each base angle is 140°.

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

4. Triangle inequality

ORIGINAL PROBLEM

Can segments of length 4, 5 and 10 form a triangle?

INCORRECT WORKED SOLUTION

5 + 10 > 4 ✓

So yes, they form a triangle.

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

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5. Possible third sides

ORIGINAL PROBLEM

Two sides of a triangle are 7 and 12. What lengths are possible for the third side?

INCORRECT WORKED SOLUTION

$7 + 12 = 19$

$12 - 7 = 5$

The third side is 5 or 19.

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

6. Congruence shortcuts

ORIGINAL PROBLEM

$\triangle ABC$ and $\triangle DEF$ have $AB \cong DE$, $BC \cong EF$ and $\angle A \cong \angle D$. Are the triangles congruent?

INCORRECT WORKED SOLUTION

Two pairs of sides and a pair of angles are congruent.

The triangles are congruent by SSA.

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

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7. Corresponding parts

ORIGINAL PROBLEM

△PQR ≅ △XYZ. XZ = 9, YZ = 15 and QR = 2x + 3. Find x.

INCORRECT WORKED SOLUTION

QR ≅ XZ

2x + 3 = 9

x = 3

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

8. CPCTC

ORIGINAL PROBLEM

Given: M is the midpoint of AD and of BC.
Prove: AB ≅ DC.

INCORRECT WORKED SOLUTION

1. AM ≅ DM (Definition of midpoint)
2. BM ≅ CM (Definition of midpoint)
3. AB ≅ DC (CPCTC)

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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9. AAA

ORIGINAL PROBLEM

In △ABC and △DEF, ∠A ≅ ∠D, ∠B ≅ ∠E and ∠C ≅ ∠F. Are the triangles congruent?

INCORRECT WORKED SOLUTION

All three pairs of angles are congruent.

The triangles are congruent by AAA.

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

10. Isosceles perimeter

ORIGINAL PROBLEM

An isosceles triangle has base 8 and perimeter 30. Find the length of each leg.

INCORRECT WORKED SOLUTION

Legs = 30 − 8

Each leg = 22

A. EXPLAIN THE MISTAKE (1 POINT)

Describe exactly which step is wrong and why.

B. SHOW CORRECT WORK (1 POINT)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points